

DANIEL HAENELT

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EDUCATION & POSITIONS

Athinoula A. Martinos Center for Biomedical Imaging

Postdoctoral Research Fellow

2024 - Present

Charlestown, MA USA

- Research group: Magnetic Resonance — Physics & Instrumentation Group
- Supervisors: Jonathan R. Polimeni and Fuyixue Wang

Max Planck Institute for Human and Cognitive and Brain Sciences

Guest Researcher

2023 - Present

Leipzig, Germany

- Research group: Department of Neurophysics

Max Planck Institute for Human and Cognitive and Brain Sciences

PhD

2017 - 2024

Leipzig, Germany

- Graduate school: International Max Planck Research School on Neuroscience of Communication: Function, Structure, and Plasticity (IMPRS NeuroCom)
- Research topic: High resolution functional magnetic resonance imaging methods at ultra-high magnetic field strength (7 T)
- Supervisor: Prof. Dr. Nikolaus Weiskopf (Department of Neurophysics)
- Final grade: summa cum laude

Leipzig University

Master of Science (M.Sc.) in Physics

2015 - 2017

Leipzig, Germany

- Master thesis: *Layer-dependent fMRI of the primary motor cortex at 7 T indicates whether we act or imitate* (Grade: 1.7)
- Final grade: 1.9

Leipzig University

Bachelor of Arts (B.A.) in Musicology and Philosophy

2012 - 2016

Leipzig, Germany

- Bachelor thesis: *Investigation of a correlation between spontaneous motor tempo and individual alpha-band brain activity* (Grade: 1.0)
- Final grade: 1.4

Free University

Bachelor of Science (B.Sc.) in Physics

2008 - 2013

Berlin, Germany

- Bachelor thesis: *Femtosekundenspektroskopie an relevanten Molekülen des interstellaren Mediums* (Grade: 1.0)
- Final grade: 2.6

German Abitur (A-level equivalent)

Goethe Oberschule

2008

Berlin, Germany

- Final grade: 1.8

EXPERIENCE

Max Planck Institute for Biological Cybernetics 06/2019 - 12/2020
Visiting researcher Tuebingen, Germany

- Research group: Department for High-field Magnetic Resonance
- Topic: Mapping ocular dominance columns in humans using fMRI at 7 T and 9.4 T

Max Planck Institute for Human and Cognitive and Brain Sciences 01/2015 - 09/2015
Student assistant Leipzig, Germany

- Research group: Department of Neurophysics
- Topic: Development of a pipeline for co-registration of high resolution functional MRI data with MIPAV using CBS High-Res Brain Processing Tools

Max Planck Institute for Human and Cognitive and Brain Sciences 12/2013 - 11/2014
Internship Leipzig, Germany

- Research group: Auditory Cognition
- Topic: Examination of resting-state EEG measurements and self-paced tapping (SMT)

Leibniz Institute for Tropospheric Research (TROPOS) 06/2012 - 12/2012
Student assistant Leipzig, Germany

- Research group: Optical Remote Sensing
- Topic: Measurement of aerosol-cloud interactions with a light detection and ranging system (LIDAR)

Institute of Scientific and Technical Research for Defense (CITEDEF) 03/2013 - 04/2013
Internship Buenos Aires, Argentina

- Research institute: Laser Institute for Atmospheric Research (CEILAP)
- Topic: Investigation of air pollution in Buenos Aires using a Light detection and ranging system (LIDAR)

University of Granada 10/2010 - 02/2011
Erasmus scholarship Granada, Spain

- Faculty abroad: Physics and Environmental Sciences

AWARDS & PRIZES

- **Magna Cum Laude Merit Award** (2020). Mapping ocular dominance columns in humans using GE-EPI, SE-EPI and SS-SI-VASO at 7 T. ISMRM & SMRT Virtual Conference & Exhibition, International Society for Magnetic Resonance in Medicine (ISMRM).
- **Magna Cum Laude Merit Award** (2026). Modeling of large-vessel BOLD contributions in Echo-Planar Time-resolved Imaging (EPTI) using subject-specific MR angiography. ISMRM & ISMRT Annual Meeting 2026, International Society for Magnetic Resonance in Medicine (ISMRM).
- **Top Trainee Abstract Selection** (2026). Modeling of large-vessel BOLD contributions in Echo-Planar Time-resolved Imaging (EPTI) using subject-specific MR angiography. ISMRM & ISMRT Annual Meeting 2026, ISMRM Brain Function Study Group.

Symposia

- **Co-Organizer of symposium** Brain in Depth (BID) 2019: International symposium on layer-dependent MRI, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.

Ad hoc reviewer

Frontiers in Neuroscience, Frontiers in Neuroanatomy, Frontiers in Neuroinformatics, Cerebral Cortex, Imaging Neuroscience, eLife, Communications Engineering (Nature Portfolio), Communications Biology (Nature Portfolio), Discover Imaging.

ARTICLES

- Stroh, A.-L., Edwards, L. J., **Haenelt, D.**, Attar, F. M., Pine, K. J., Trampel, R., Szwed, M., Weiskopf, N. (2026). Congenital blindness reduces myelination in human visual cortex. *Science Advances*, 12, eaec2348.
- **Haenelt, D.**, Chaimow, D., Schmidt, M. E., Nasr, S., Weiskopf, N., Trampel, R. (2026). Decoding of columnar-level organization across cortical depth using BOLD- and CBV-fMRI at 7 T. *Imaging Neuroscience*, 4, IMAG.a.1124.
- **Haenelt, D.**, Trampel, R., Chaimow, D., Shmuel, A., Sereno, M. I., Weiskopf, N. (2025). Spatial specificity of the functional gradient echo and spin echo BOLD signal across cortical depth at 7 T. *bioRxiv*, 1–33.
- Stroh, A.-L., **Haenelt, D.**, Edwards, L. J., Attar, F. M., Pine, K. J., Trampel, R., Szwed, M., Weiskopf, N. (2025). Myelination of the early visual cortex depends on sensory experience during development. *International Journal of Psychophysiology*, 213 (Supplement), 113109.
- Chaimow, D., Degutis, J. K., **Haenelt, D.**, Trampel, R., Weiskopf, N., Lorenz, R. (2026). Re-evaluating laminar specificity of working memory in human prefrontal cortex. *bioRxiv*, 1–12.
- Attar, F. M., Kirilina, E., Chaimow, D., **Haenelt, D.**, Schneider, C., Edwards, E. L., Pine, K. J., Jäger, C., Reimann, K., Pohlmann, A., Periquito, J., Streubel, T., Trampel, R., Mohammadi, S., Niendorf, T., Morawski, M., Weiskopf, N. (2025). Short association fibres form topographic sheets in the human V1-V2 processing stream. *Imaging Neuroscience*, 3, imag_a_00498.
- Degutis, J. K., Chaimow, D., **Haenelt, D.**, Assem, M., Duncan, J., Haynes, J.-D., Weiskopf, N., Lorenz, R. (2024). Dynamic layer-specific processing in the prefrontal cortex during working memory. *Communications Biology*, 7(1140), 1–11.
- Attar, F. M., Kirilina, E., **Haenelt, D.**, Trampel, R., Pine, K. J., Edwards, L. J., Weiskopf, N. (2024). Mapping short association fibre connectivity up to V3 in the human brain *in vivo*. *Cerebral Cortex*, 34(7), bhae279.
- Attar, F. M., Kirilina, E., Edwards, L. J., **Haenelt, D.**, Pine, K. J., Trampel, R., Chaimow, D., Weiskopf, N. (2024). Connectivity at fine scale: mapping structural connective fields by tractography of short association fibres *in vivo*. *bioRxiv*, 1–31.

- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Pine, K. J., Edwards, L. J., Helbling, S., Weiskopf, N. (2023). High-resolution quantitative and functional MRI indicate lower myelination of thin and thick stripes in human secondary visual cortex. *eLife*, 12, e78756.
- Iamshchinina, P., **Haenelt, D.**, Trampel, R., Weiskopf, N., Kaiser, D., Cichy, R. M. (2021). Benchmarking GE-BOLD, SE-BOLD, and SS-SI-VASO sequences for depth-dependent separation of feed-forward and feedback signals in high-field MRI. *bioRxiv*, 1–18.
- Iamshchinina, P., Kaiser, D., Yakupov, R., **Haenelt, D.**, Sciarra, A., Mattern, H., Luesebrink, F., Duezel, E., Speck, O., Weiskopf, N., Cichy, R. M. (2021). Perceived and mentally rotated contents are differentially represented in cortical depth of V1. *Communications Biology*, 4(1069), 1–8.
- Attar, F. M., Kirilina, E., **Haenelt, D.**, Pine, K. J., Trampel, R., Edwards, L. J., Weiskopf, N. (2020). Mapping short association fibers in the early cortical visual processing stream using in vivo diffusion tractography. *Cerebral Cortex*, 30(8), 4496–4514.

TALKS AND POSTER PRESENTATIONS

- Nguyen, N., **Haenelt, D.**, Tran, D.-D., Cao, T., Trampel, R., Weiskopf, N. (2026). Laminar segregation of slow hemodynamic oscillations in human V1 via geometric and dynamic mode analysis. Poster presented at the 32th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Bordeaux, France.
- **Haenelt, D.**, Hartung, G., Berman, A., Chen, J., Wu, J., Bilgic, B., Frost, R., Dong, Z., Wang, F., Polimeni, J. R. (2026). Modeling of large-vessel BOLD contributions in Echo-Planar Time-resolved Imaging (EPTI) using subject-specific MR angiography. Talk at ISMRM & ISMRT Annual Meeting 2026, Cape Town, South Africa.
- **Haenelt, D.**, Dahmani, L., Wu, J., Liu, H., Polimeni, J. R., Dong, Z., Wang, F. (2026). Deep contrastive variational autoencoder for physiological noise removal of Echo-Planar Time-resolved (EPTI) fMRI. Talk at ISMRM & ISMRT Annual Meeting 2026, Cape Town, South Africa.
- Hartung, G., Berman, A., **Haenelt, D.**, Schillinger, D., Polimeni, J. R. (2026). Modeling effects of EPI readout on laminar profiles of spin-echo BOLD via biophysical simulations using realistic vasculature. Talk at ISMRM & ISMRT Annual Meeting 2026, Cape Town, South Africa.
- Gopinath, K., Wightton, P., Balasubramanian, M., Frost, R., Huber, R., Droppa, K., Greve, D. N., Nielsen, J. D., Puonti, O., **Haenelt, D.**, Van der Kouwe, A. J., Fischl, B., Polimeni, J. R., Iglesias, J. E., Varadarajan, D. (2026). SurfScribe: Cortical surface-driven automated online slice prescription applied to ultra-high-resolution vascular MRI. Poster presented at ISMRM & ISMRT Annual Meeting 2026, Cape Town, South Africa.
- **Haenelt, D.**, Dahmani, L., Wu, J., Liu, H., Polimeni, J. R., Dong, Z., Wang, F. (2026). Deep contrastive variational autoencoder for physiological noise removal of Echo-Planar Time-resolved (EPTI) fMRI. Poster presented at MIT Medical Imaging Initiative Symposium, Cambridge, MA, USA.
- **Haenelt, D.**, Polimeni, J. R., Chang, Y., Bollmann, S., Gomez, D. E. P., Abbasi-Rad, S., Van der Kouwe, A., Frost, R. (2025). Volumetric navigators for prospective motion correction of 0.16-mm isotropic time-of-flight angiography at 7 T. Poster presented at the 7T Translational Alliance in North America (7T TANA) Meeting 2025, Dallas, TX, USA.

- Dong, Z., **Haenelt, D.**, Nasr, S., Rosen, B. R., Wald, L. L., Polimeni, J. R., Wang, F. (2025). Mesoscale fMRI at ultra-high spatiotemporal resolutions using 3D Echo Planar Time and DYnamic-resolved Imaging (TIDY). Talk at ISMRM & ISMRT Annual Meeting 2025, Honolulu, HI USA.
- **Haenelt, D.**, Polimeni, J. R., Chang, Y., Bollmann, S., Gomez, D. E. P., Abbasi-Rad, S., Van der Kouwe, A., Frost, R. (2025). High-resolution 0.16-mm time-of-flight angiography at 7 T with volumetric navigators for prospective motion correction. Talk at ISMRM & ISMRT Annual Meeting 2025, Honolulu, HI USA.
- **Haenelt, D.**, Dong, Z., Hu, Z., Setsompop, K., Wang, F., Polimeni, J. R. (2025). Comparing BOLD contamination in CBV-based fMRI with conventional EPI and echo-planar time-resolved imaging (EPTI) at 7 T. Poster presented at ISMRM & ISMRT Annual Meeting 2025, Honolulu, HI USA.
- **Haenelt, D.**, Polimeni, J. R., Chang, Y., Bollmann, S., Gomez, D. E. P., Abbasi-Rad, S., Van der Kouwe, A., Frost, R. (2025). Volumetric navigators for prospective motion correction of 0.16-mm isotropic time-of-flight angiography at 7 T. Poster presented at ISMRM Joint Workshop on Ultrahigh Field Magnetic Resonance & Brain Function, Annapolis, MD USA.
- Stroh, A.-L., **Haenelt, D.**, Edwards, L. J., Attar, F. M., Pine, K. J., Trampel, R., Szwed, M., Weiskopf, N. (2024). Zooming in: Using ultra-high resolution quantitative and diffusion MRI to uncover the underlying microstructural mechanisms of structural plasticity in congenitally blind humans. Poster presented at the Seeing and Acting Workshop (SAW): Functional and Neural Perspectives, Coimbra, Portugal.
- Schmidt, M. E., **Haenelt, D.**, Kirilina, E., Weiskopf, N. (2024). Enhanced specificity for single eye ocular dominance column (ODC) mapping using neurophysiologically informed spatial filtering in ultra-high-resolution fMRI. Poster presented at the annual meeting of the Society for Neuroscience (SfN), Chicago, IL USA.
- **Haenelt, D.**, Polimeni, J. R., Chang, Y., Gomez, D. E. P., Van der Kouwe, A., Frost, R. (2024). Volumetric navigators for prospective motion correction of slab-selective high-resolution GRE at 7 T. Poster presented at ISMRM Workshop on Motion Correction in MR, Québec City, Canada.
- Jastrzębowska, M. A., Gravel, N., Iamshchinina, P., **Haenelt, D.**, Weiskopf, N., Cichy, R. M. (2024). Cortical depth-dependent population receptive field size variation in human V1, V2 and V3. *Journal of Vision*, 24(10), 846.
- Stroh, A.-L., **Haenelt, D.**, Edwards, L. J., Attar, F. M., Pine, K. J., Trampel, R., Szwed, M., Weiskopf, N. (2024). Zooming in: Using ultra-high resolution quantitative and diffusion MRI to uncover the underlying microstructural mechanisms of structural plasticity in congenitally blind humans. Poster presented at the 22nd annual International Multisensory Research Forum (IMRF), Reno, NV USA.
- Chaimow, D., Degutis, J. K., **Haenelt, D.**, Trampel, R., Weiskopf, N., Lorenz, R. (2024). Laminar fMRI in the human prefrontal cortex at 7 T: A replication study of layer-dependent activity during working memory. Poster presented at the Brain in Depth (BID) Conference 2024, Tuebingen, Germany.
- Jastrzębowska, M. A., Gravel, N., Iamshchinina, P., **Haenelt, D.**, Weiskopf, N., Cichy, R. M. (2023). Population receptive field sizes in human V1, V2 and V3 varies as a function of cortical depth. Poster presented at the European Conference on Visual Perception (ECVP), Paphos, Cyprus.

- **Haenelt, D.**, Weiskopf, N., Chaimow, D., Nasr, S., Trampel, R. (2023). Cortical depth-dependence of ocularity decoding in high-resolution fMRI at 7 T. Poster presented at the 29th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Montreal, Canada.
- **Haenelt, D.**, Trampel, R., Chaimow, D., Sereno, M. I., Weiskopf, N. (2022). Examination of draining vein contributions in GE- and SE-EPI BOLD across cortical depth at 7T. Poster presented at 28th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Glasgow, Scotland.
- Chaimow, D., Degutis, J. K., **Haenelt, D.**, Trampel, R., Weiskopf, N., Lorenz, R. (2022). Towards a reproducible layer fMRI pipeline for high-level cognition. Poster presented at 28th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Glasgow, Scotland.
- Degutis, J. K., Chaimow, D., **Haenelt, D.**, Assem, M., Duncan, J., Weiskopf, N., Lorenz, R. (2022). Increased working memory load activates superficial layers of the dorsolateral prefrontal cortex. Poster presented at 28th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Glasgow, Scotland.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Pine, K. J., Edwards, L. J., Helbling, S., Weiskopf, N. (2021). Myelination differences of stripes in human V2: preliminary evidence from 7 T quantitative MRI. Poster presented at 27th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Virtual Conference.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Pine, K. J., Edwards, L. J., Helbling, S., Weiskopf, N. (2021). Myelination differences of stripes in human V2: preliminary evidence from 7 T quantitative MRI. Poster presented at Brain in Depth (BID) 2021: Annual Meeting on Meso-scale Imaging, Virtual Meeting.
- Iamshchinina, P., Kaiser, D., Yakupov, R., **Haenelt, D.**, Sciarra, A., Mattern, H., Duezel, E., Speck, O., Weiskopf, N., Cichy, R. M. (2020). Perceived and mentally rotated contents are differentially represented in cortical layers of V1. Poster presented at 26th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Virtual Conference.
- **Haenelt, D.**, Weiskopf, N., Vaculciakova, L., Mueller, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Huber, L., Sereno, M. I., Trampel, R. (2020). Mapping ocular dominance columns in humans using GE-EPI, SE-EPI and SS-SI-VASO at 7 T. Talk at ISMRM & SMRT Virtual Conference & Exhibition 2020, Virtual Conference.
- **Haenelt, D.**, Weiskopf, N., Mueller, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Trampel, R. (2019). Reliable 3D mapping of ocular dominance columns in humans using GE-EPI fMRI at 7 T. Poster presented at 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Attar, F. M., Kirilina, E., **Haenelt, D.**, Trampel, R., Edwards, L. J., Weiskopf, N. (2019). Mapping visual field specific short V1-V2 connectivity using high resolution dMRI and fMRI. Poster presented at 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Rome, Italy.
- **Haenelt, D.**, Weiskopf, N., Mueller, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Trampel, R. (2019). Reliable 3D mapping of ocular dominance columns in humans using GE-EPI fMRI at 7 T. Poster presented at Brain in Depth (BID) 2019: International Symposium on layer-dependent MRI, Leipzig, Germany.

- Attar, F. M., Kirilina, E., **Haenelt, D.**, Edwards, L. J., Pine, K. J., Weiskopf, N. (2019). Mapping short association fibres in the human visual system with ultra high resolution and high sensitivity diffusion MRI. Talk at ISMRM & SMRT Virtual Conference & Exhibition 2019, Montreal, Canada.
- Neef, N. E., Trampel, R., Lu, L., **Haenelt, D.**, Bazin, P.-L., Turner, R., Weiskopf, N., Friederici, A. D. (2019). Imagined and actual speaking disentangle involvement of motor and somatosensory cortices: submillimeter resolution resolves the cortical organization of speech pronunciation. Poster presented at Symposium: Neural Bases of Speech Production, San Francisco, CA USA.
- **Haenelt, D.**, Trampel, R., Mueller, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Weiskopf, N. (2019). Reliable mapping of columnar structures in early visual cortex using GE-EPI at 7 T. Poster presented at ISMRM Workshop on Ultrahigh Field Magnetic Resonance, Dubrovnik, Croatia.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Weiskopf, N. (2018). Mapping colour-selective columns in V2 across cortical depth using GE- and SE-EPI. Poster presented at Interpreting BOLD: Furthering the dialogue between cognitive and cellular neuroscience, Oxford, UK.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Weiskopf, N. (2018). Mapping colour-selective columns in V2 across cortical depth using GE- and SE-EPI. Poster presented at 24th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Singapore.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni, J. R., Tootell, R. B. H., Sereno, M. I., Weiskopf, N. (2018). Mapping colour-selective columns in V2 across cortical depth using GE- and SE-EPI. Poster presented at Brain in Depth (BID) 2018: Symposium on layer-dependent MRI, Magdeburg, Germany.
- **Haenelt, D.**, Trampel, R., Bazin, P.-L., Weiskopf, N., Kuehn, E. (2017). Layer-dependent BOLD of M1 during observed and actual finger tapping. Poster presented at the 7th summer school of the International Max Planck Research School on Neuroscience of Communication: Function, Structure, and Plasticity (IMPRS NeuroCom), London, UK.

PUBLISHED SOFTWARE

- **Haenelt, D.** (2020). GBB — Gradient-based boundary: A Python package to improve registration accuracy between anatomical and functional MR data (0.1.6), <https://pypi.org/project/gbb/>.
- **Haenelt, D.** (2024). FmriTools (1.0.3). Zenodo, <https://doi.org/10.5281/zenodo.10573042>.
- **Haenelt, D.** (2026). synthBOLD (1.0.0). Zenodo, <https://doi.org/10.5281/zenodo.21708862>.
- **Haenelt, D.** (2026). epti2bids — A Docker container to convert EPTI and ME-EPI fMRI data to BIDS format (1.0.0), <https://hub.docker.com/r/haenelt/epti2bids>.

OPEN SCIENCE

- **Haenelt, D.**, Trampel, R., Chaimow, D., Shmuel, A., Sereno, M. I., Weiskopf, N. (2025). Spatial specificity of the functional gradient echo and spin echo signal across cortical depth at 7 T, Open dataset, <https://osf.io/fh7pr/>.

- **Haenelt, D.**, Chaimow, D., Nasr, S., Weiskopf, N., Trampel, R. (2023). Decoding of columnar-level organization across cortical depth using BOLD- and CBV-fMRI at 7 T. Open dataset, <https://osf.io/awmc6/>.
- **Haenelt, D.**, Trampel, R., Nasr, S., Polimeni J. R., Tootell, R. B. H., Sereno, M. I., Pine, K. J., Edwards, L. J., Helbling, S., Weiskopf, N. (2022). High resolution quantitative and functional MRI indicate lower myelination of thin and thick stripes in human secondary visual cortex, Open dataset, <https://osf.io/624cz/>.

SKILLS

Programming	Python, Matlab, Rust, Bash, L ^A T _E X
Neuroimaging software	FreeSurfer, SPM, FSL, ANTs, AFNI
Behavioral software	PsychoPy [®] , Psychtoolbox-3, Presentation [®] (Neurobehavioral Systems)
3D software	Blender
Languages	German (native), English (fluent), French (advanced), Spanish (advanced)